ICS 4111: Embedded Systems & IoT
Practical Exercise 2 – Part A Calculations

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CIRCUIT A – Series Circuit (125 V | 20 Ω – 30 Ω – 50 Ω)

1. Total Resistance

In a series circuit, total resistance is the sum of all individual resistances:

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R_total = R1 + R2 + R3
R_total = 20 + 30 + 50
```

R_total = 100 Ω

2. Total Current (Ohm's Law: I = V / R)

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I_total = V / R_total = 125 / 100
```

I_total = 1.25 A

In a series circuit the same current flows through every component.

3. Current Through Each Resistor

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I_R1 = I_R2 = I_R3 = 1.25 A (series circuit – same current everywhere)
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4. Voltage Drop Across Each Resistor (V = I × R)

Resistor	Value	Current	Voltage Drop	Check
R1	20 Ω	1.25 A	V = 1.25 × 20 = 25.00 V	✓
R2	30 Ω	1.25 A	V = 1.25 × 30 = 37.50 V	✓
R3	50 Ω	1.25 A	V = 1.25 × 50 = 62.50 V	✓
Total	**100 Ω**	**1.25 A**	**25 + 37.5 + 62.5 = 125 V**	**= V_supply ✓**

5. Power Dissipated (Watt's Law: P = V × I)

Resistor	Voltage Drop	Current	Power (P = V × I)
R1	25.00 V	1.25 A	P = 25.00 × 1.25 = 31.25 W
R2	37.50 V	1.25 A	P = 37.50 × 1.25 = 46.875 W
R3	62.50 V	1.25 A	P = 62.50 × 1.25 = 78.125 W
Total	**125 V**	**1.25 A**	**P_total = 125 × 1.25 = 156.25 W**

CIRCUIT B – Parallel Circuit (125 V | 20 Ω ■ 100 Ω ■ 50 Ω)

1. Total Resistance

For parallel resistors: 1/R_total = 1/R1 + 1/R2 + 1/R3

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1/R_total = 1/20 + 1/100 + 1/50
1/R_total = 5/100 + 1/100 + 2/100 = 8/100
```

R_total = 100/8 = 12.5 Ω

2. Total Current

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I_total = V / R_total = 125 / 12.5
```

I_total = 10.0 A

3. Current Through Each Resistor (I = V / R, each sees full 125 V)

Resistor	Value	Voltage	Current (I = V/R)
R1	20 Ω	125 V	I = 125 / 20 = 6.250 A
R2	100 Ω	125 V	I = 125 / 100 = 1.250 A
R3	50 Ω	125 V	I = 125 / 50 = 2.500 A
Total	**12.5 Ω**	**125 V**	**I = 6.25+1.25+2.5 = 10.00 A ✓**

4. Voltage Drop Across Each Resistor

In a parallel circuit every branch is directly connected across the supply:

V_R1 = V_R2 = V_R3 = 125 V (equal to supply voltage)

5. Power Dissipated (P = V × I)

Resistor	Voltage	Current	Power (P = V × I)
R1	125 V	6.250 A	P = 125 × 6.250 = 781.250 W
R2	125 V	1.250 A	P = 125 × 1.250 = 156.250 W
R3	125 V	2.500 A	P = 125 × 2.500 = 312.500 W
Total	**125 V**	**10.00 A**	**P_total = 125 × 10 = 1250.000 W**

Summary Table

Parameter	Circuit A (Series)	Circuit B (Parallel)
Supply Voltage	125 V	125 V
Total Resistance	100 Ω	12.5 Ω
Total Current	1.25 A	10.00 A
Voltage across R1	25.00 V	125 V
Voltage across R2	37.50 V	125 V
Voltage across R3	62.50 V	125 V
Current through R1	1.25 A	6.25 A
Current through R2	1.25 A	1.25 A
Current through R3	1.25 A	2.50 A
Power in R1	31.25 W	781.25 W
Power in R2	46.875 W	156.25 W
Power in R3	78.125 W	312.50 W
Total Power	156.25 W	1250.00 W